ECE 592 Homework 2 Cache Reverse Engineering

Measurement controls, side-channel attacks, and data-memory-dependent prefetching

01 Measure

02 Attack

03 Investigate

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Controls determine what timing can tell us

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HW1 baselines reviewed

Five completed measurement lessons

01 Comparable time
Convert machine ticks to nanoseconds.

02 Cache boundaries
Sweep densely around transitions, including non-power-of-two capacities.

03 Inclusion
Separate the eviction and probe cores. Use a decoy set to test causality.

04 Page placement and translation
Vary page setups and compare counters to separate cache, placement, and translation effects.

05 Associativity
Use latency transitions when eviction probabilities are too noisy.

Cache behavior depends on workload

H1	**Direct-mapped caches**	**Contradicted in the reviewed results** Measured L1 caches remained set-associative while hit latency stayed roughly flat.
H2	**The memory wall**	**Supported as workload dependent** Sequential access stayed faster than random access at large footprints. Bandwidth untested.
H3	**Stream buffers**	**Benefit depends on access predictability** Sequential gains may weaken when pointer chasing makes future addresses unpredictable.

More metadata, a smaller share of the cache

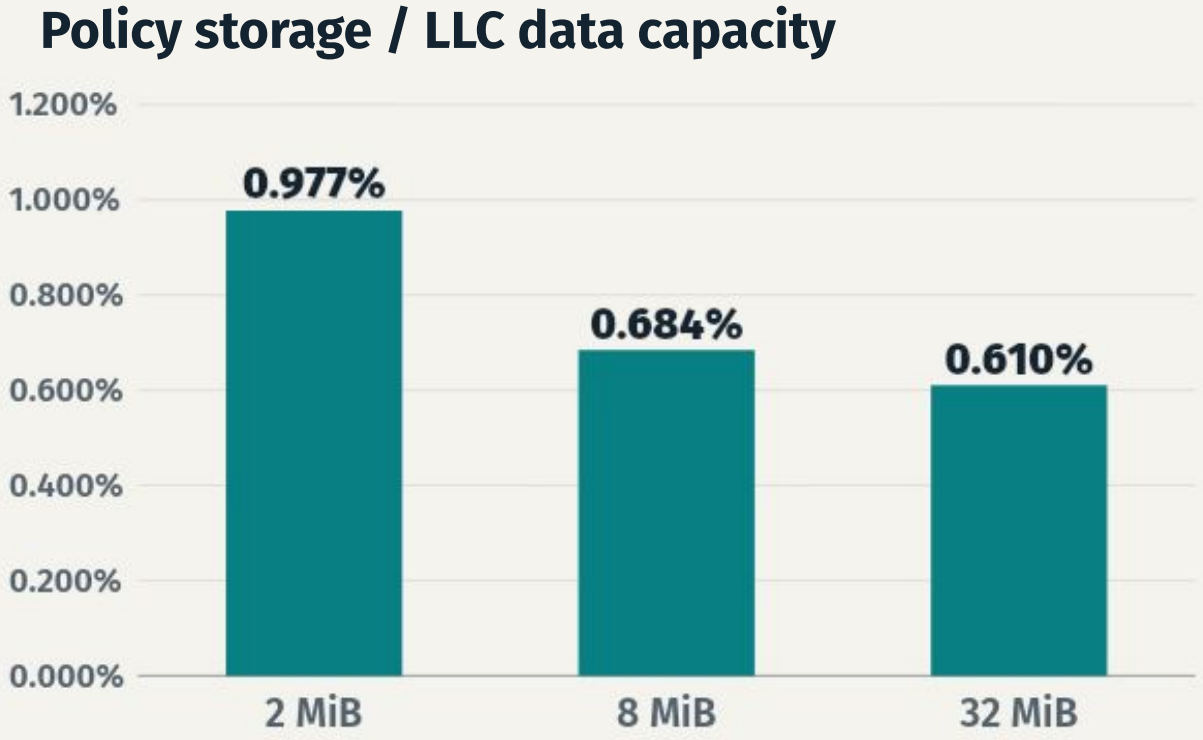
20 KiB

at a 2 MiB LLC

200 KiB

at a 32 MiB LLC

3 bits per line + 8 KiB fixed



Calculation assumes fixed predictor and sample sizes.
It does not establish a speedup or energy saving.

Three ways to read cache activity

METHOD	WHAT THE ATTACKER TIMES	EXPERIMENT
Prime+Probe	Reloads of two primed L1 sets after a simulated victim access	Same-process secret-bit test 100,000 main trials per host
Flush+Reload	A reload after the benchmark caches or evicts a line	Randomized hit / miss test 20,000 trials per host
Flush+Flush	The flush instruction on a cached or uncached line	Single and split-thread variants Usually 10,000 test trials

Six hosts separated the secret bit clearly

94.1–99.9% on six hosts



Skylark: 51.842% accuracy, near the 50% chance level

Thunderbird unsupported by this x86 code. Greenforest and Elevation were outside this run.

Greenforest gave the clearest hit / miss separation

99.98%

Greenforest accuracy



Greenforest: 22.21 ticks for a hit, 153.51 for a miss

Other hosts reached 75.10–81.83% accuracy. Each host used 20,000 randomized trials.

Warm-up changed the reported results



+10k

warm-up samples

Cold timings were about 5–6× higher.

Thread split still mattered: 48.4–80.7% accuracy across hosts

Elevation’s rerun also used more test samples. These are the reported before / after runs.

Defense must match the source of leakage

DEFENSE	PRIME+PROBE	FLUSH+RELOAD	MAIN COSTS
CEASER	Changing mappings disrupts learned conflicts	Shared-line timing remains possible	Encryption logic, remapping
ScatterCache	Secret mappings depend on domain and cache way	Separate process IDs eliminate timing analysis	Mapping logic, address metadata

An LLC-only defense would not directly stop attacks on L1

Separating cached data can still leave shared replacement state exposed.

Can an unread pointer bring its target into cache?

20,000 trials per condition / 4 MB target pool / Pinned core

64-entry pointer array

Entries 0–31	Entries 32–63
Read as integers	Leave unread

1 **Flush all targets**

2 **Read 32 entries**

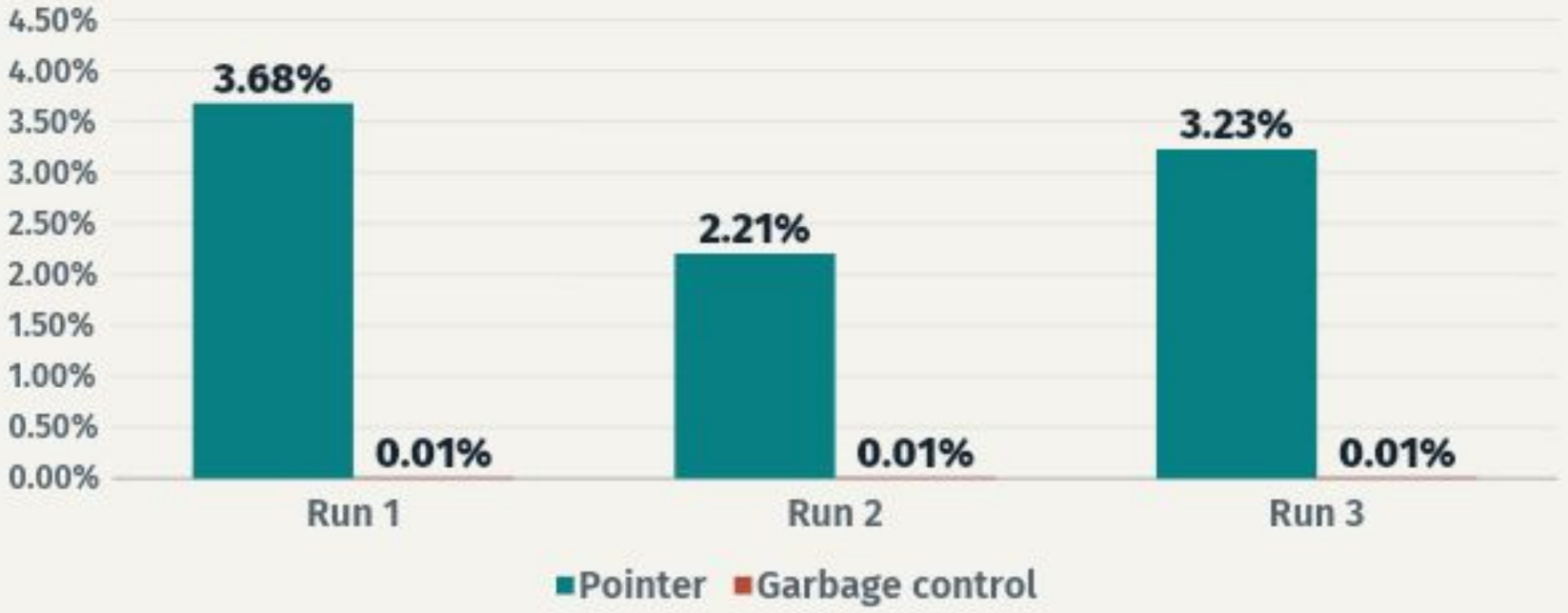
3 **Wait 1,000 iterations**

4 **Time an unread target**

CONTROL Change only the probed array entry to a non-pointer value. Keep the addresses, flushes, and access pattern identical.

A repeatable signal on Greenforest

Overall cache-hit rate after flush



26–50%

peak hit rate at distance 1

i7-14700
Raptor Lake

DMP observed on Greenforest under this test

Seven other machines: not observed under tested conditions. Absence remains unproven.

What the experiments establish

01 Controls shape the conclusion

Calibration, warm-up, and decoy tests determine whether a timing difference supports the intended claim.

02 Leakage varies by machine and setup

Prime+Probe was strong on six hosts. Flush-based results changed substantially with hardware and conditions.

03 Memory contents can change cache state

Greenforest repeatedly cached flushed, unread targets. The tested pattern supports DMP evidence with clear limits.